

COMPACT ROBOTIC PLATFORM With Motor Driver On Chassis PCB

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An N20 motor is a compact DC geared motor widely used in robotics and DIY applications. By combining a small DC motor with an integrated gearbox, it delivers high torque in a compact form factor. Motor drivers are essential components in robotic systems, enabling precise control of motor speed and direction. Integrating motor-driver circuitry and sensor interfaces directly onto the robot chassis simplifies construction, reduces wiring complexity, and improves overall reliability.

The proposed system integrates the motor-driver circuitry directly into the robot chassis PCB, creating a compact, self-contained robotic platform. It enables precise control of the robot's direction, speed, and individual wheel movement through a single FPC (flexible printed circuit) cable, simplifying assembly and reducing the number of interconnections.

The system supports various control methods, including Wi-Fi and other communication interfaces. Conventional robot designs often require separate connections for motor drivers, batteries, displays, sensors, and microcontrollers, leading to complex wiring and longer assembly times.

To overcome these limitations, a compact robot PCB has been developed that serves as both the main circuit board and the structural chassis. N20 motors can be mounted directly onto the PCB alongside the motor driver, battery interface, power management circuitry, and display connections. The board supports surface-mount assembly and includes provisions

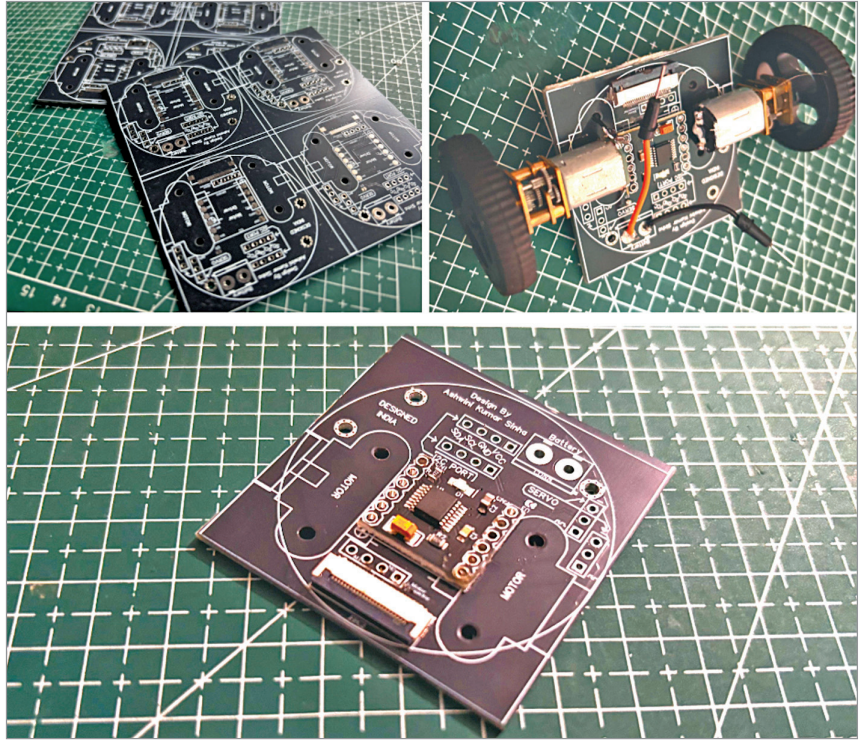


Fig. 1: Author's prototype and developed PCB

BILL OF MATERIALS

Component/Part Description	Designator	Package/Footprint	Quantity
10 μ F capacitor	C1	C0805	1
2.2 μ F capacitor	C2	C0805	1
10nF capacitor	C3	C0805	1
DRV8833PW dual motor driver IC	U1	HTSSOP-16 (5.0mm $\times$ 4.4mm, 0.65mm pitch)	1
DRV8833 dual motor driver module	U2	DRV8833 motor driver module	1
2-pin screw terminal	CON1	Screw_terminal_2PIN	1
24-pin FPC connector (XW0520012-240R-001)	FPC1	FPC-SMD, 24P, 0.5mm pitch	1
4-pin header	H1, H2, H5	Header_4PIN_VGT	3
3-pin header	H3, H4	Header_3PIN_VGT	2
N20 DC geared motor	M1,M2	N20_DC_Motor	1

for additional servo motors, further reducing wiring requirements.

Despite its compact form factor, the complete robot chassis measures only about 4cm. The PCB includes

an FPC connector and header pins for easy interfacing with external microcontrollers. When used with the IndusBoard Coin, the controller can be connected directly through a single